GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Cleaning Machine Motor Care & Inspection Schedule

For fleet managers, service technicians and OEM designers of battery floor scrubbers, scrubber-driers and sweepers

Use the shift checks as an operator routine, the monthly and seasonal checks as a technician schedule, and the calculation and specification sections when buying or designing machines. Record readings so trends show up before failures do.

Every shift — operator checks

- [ ] Wipe motor areas, decks and connectors; never pressure-wash or hose motors, connectors or the controller.
- [ ] Empty and rinse the recovery tank; check that the float shut-off moves freely and the vacuum filter is clean.
- [ ] Remove string, strapping and hair wound round brush hubs, pad drivers and wheel axles.
- [ ] Keep deck pressure at or below the set ceiling for the pad and floor in use.
- [ ] Recharge the battery fully after the shift; never leave the machine discharged.

Monthly — technician checks

- [ ] Inspect carbon brushes on brush-deck, traction and vacuum motors; replace in sets at the minimum-length mark.
- [ ] Check commutators for burning, grooving or heavy carbon deposits; blow out with dry, low-pressure air.
- [ ] Inspect cable glands, connectors and terminals for corrosion or detergent residue.
- [ ] Confirm the parking brake releases fully at normal battery voltage; check the air gap against the brake data sheet.
- [ ] Record brush-motor current at the normal deck setting and traction current on a fixed route; compare with last month.

Before the monsoon and before storage

- [ ] Measure winding insulation resistance at 500 V DC with the controller and electronics disconnected; investigate any falling trend.
- [ ] Turn each motor and wheel by hand; rough or gritty rotation indicates bearing corrosion.
- [ ] Check shaft seals and axle seals for leaks or hardening.
- [ ] Dry the machine fully and store it in a ventilated area, not a damp store room.
- [ ] During idle periods, run each motor briefly every one to two weeks.

Overload checks — deck pressure and ramps

- [ ] Brush torque for a disc pad: $T = (2/3) \times \mu \times Fn \times R$, where Fn is down-force (N) and R is pad radius (m).
- [ ] Example: 510 mm pad, $\mu = 0.35$, $Fn = 450$ N gives 26.8 Nm; at 700 N it rises to 41.7 Nm, with about 2.4 × the copper loss.
- [ ] Traction force: $F = m \times 9.81 \times (\sin\theta + Crr \times \cos\theta)$; every 1 % of gradient adds about 1 % of machine weight.
- [ ] Example: 450 kg machine, $Crr = 0.02$ needs 88 N on the flat, 177 N on 2 % and 527 N on a 10 % ramp.
- [ ] Axle torque = F × wheel radius; compare with the transaxle's short-time axle torque rating.
- [ ] Brake holding force on a ramp = $m \times 9.81 \times \sin\theta$; about 439 N for 450 kg on 10 %.

Specification checklist for new machines

- [ ] IP rating chosen for each motor position and the real cleaning method, not the brochure.
- [ ] Contact-sealed (2RS-type) bearings and corrosion-resistant shafts wherever water can reach.
- [ ] Transaxle and parking brake sized for the steepest ramp on site with a full load and operator.
- [ ] Deck-pressure ceiling or current-based pressure control on the brush deck.
- [ ] Controller low-voltage cut-out and current limits set within the motor ratings.
- [ ] Brushless motors considered for vacuum and other high-hour duties.

Quick fault guide

Symptom	Likely cause	First action
Brush motor hot or tripping	Deck pressure too high, aggressive pad	Reduce pressure, check current
Vacuum motor short life	Water ingestion, worn brushes	Repair float, replace brushes
Stalls on a ramp	Traction overload, current limit, brake drag	Check gradient, limit and brake
Low megger reading after washing	Water ingress, motor breathing	Dry out, stop hosing
Slow late in the shift	Deep battery discharge	Charge every shift, test cells
Rough bearings after storage	Bearing corrosion	Fit sealed bearings, store dry

Intervals and values are typical guidance; follow the machine manufacturer's service manual where it differs. For sealing, transaxle sizing or brushless upgrades, contact Globe Scott Motors through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.